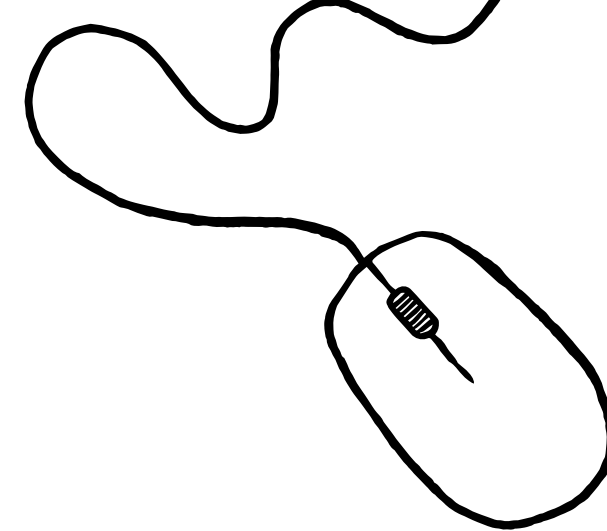




HACKATHON

Hackathon event organization and management



supervisor: Ms.Le Thi Thuy Chi





HACKATHON TEAM



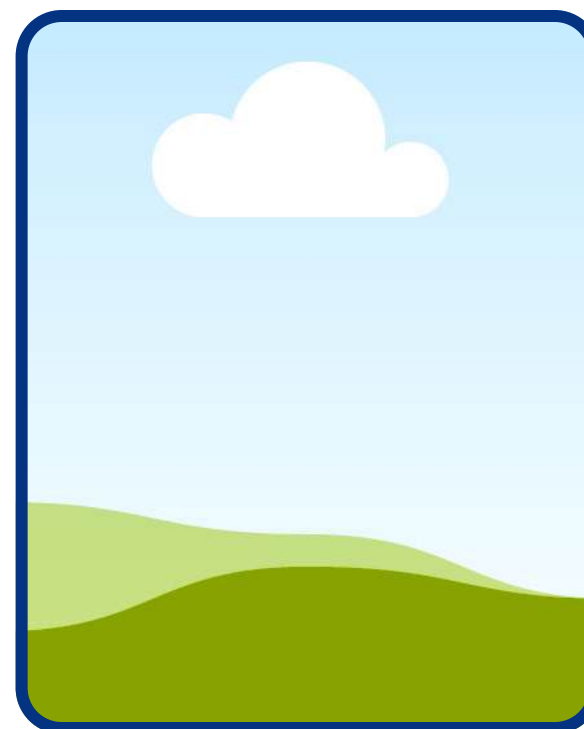
Nguyễn Trường Vinh

Menber



Trần Trúc Vỹ

Menber



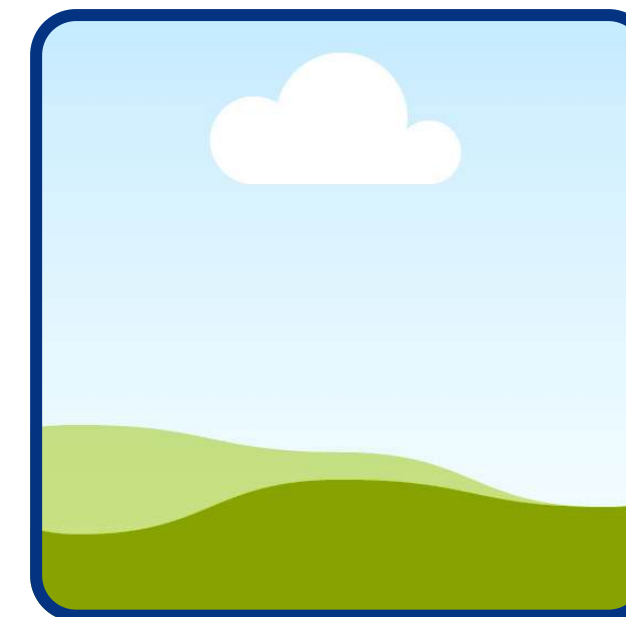
Phan Mạnh Quỳnh

Leader



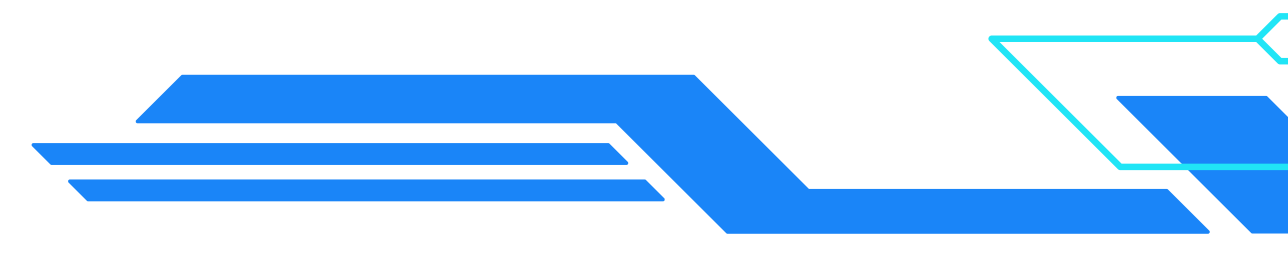
Phan Lại Phong

Menber



Phan Nguyễn Đình Khôi

Menber





AGENDA

Context

01

Proposed Solution

02

Main Actor and Features

03

Flows

04

Limitations

05

Q&A

06



1. CONTEXT

Manual Eligibility Checks

FPT students and external students enter through different doors, and every entry is verified by hand.



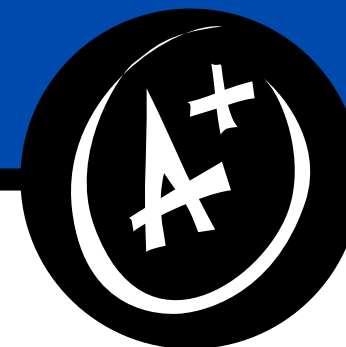
Team Formation in Chat Threads

Team size limits, invitations, join and leave requests all live in chat threads and spreadsheets.



Spreadsheet-Based Scoring

Weighted rubrics across several judges, with no conflict-of-interest check and no record of who changed what.



Results Hard to Defend

Judges disagree, ties happen at the cut-off line, and teams have no formal channel to contest a score.



2. Proposed Context

1

Event & Team Operations

- Full event lifecycle: UPCOMING → OPEN → CLOSED_REGISTRATION → ACTIVE → SCORING → COMPLETED
- Rounds, tracks, schedule and milestone submission gates
- Magic-link registration and external-student verification
- Team creation, invitations, join/leave requests, auto team matching
- Track assignment: manual, random, or self-draw

1

Defensible Scoring & Ranking

- Scoring templates whose criteria weights must total 100%
- Conflict-of-interest detection — a mentor cannot judge their own team
- Automatic deviation review when judges disagree past a threshold
- Weighted aggregation, tie-break, finalist selection and awards
- Score review and dispute workflow for teams

1

Transparency & Accountability

- Five account types with role checks enforced on every endpoint
- Append-only audit log — no update, no delete, exportable by admin
- Domain events driving e-mail and in-app notifications
- Live leaderboard, team progress tracking and risk flags
- Public event pages and published results for spectators

2. Proposed Context Tech Stack



Backend

Java Spring Boot



Database

SQL sever management
studio



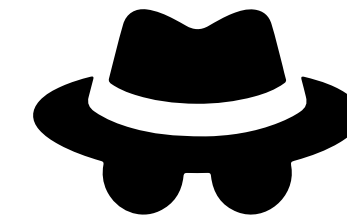
Frontend

Next JS + Node JS





3. Main Actor and Features



Guest

Feature:

- View informations of events that are active and open registration
- Register an account to view more information and regist in to an event



3. Main Actor and Features



Registered User

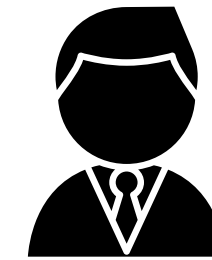
Feature:

- View informations of all events.
- Register into an event.
- Create team, finding team, invite member, team management.
- View the dashboard
- Feedback after the event ended





3. Main Actor and Features



Registered User

Feature:

Event start:

- Submit team's assignment
- View event's teams live results
- Get final result
- View certificate



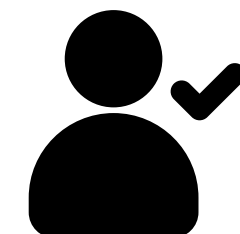
3. Main Actor and Features

Feature:

- View informations of all events.
- Score the assigned teams.
- View the dashboard



Lecture/judge



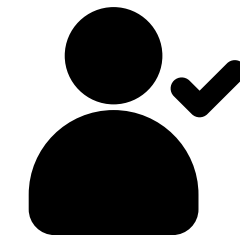
3. Main Actor and Features



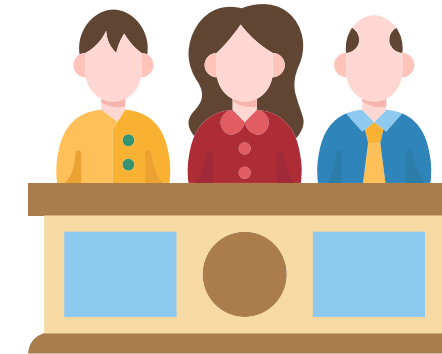
Lecture/mentor

Feature:

- View informations of all events.
- Notified about the progress of the assignments for the teams that the mentor is assigned to.
- Mentoring with the teams that the mentor is assigned to through chat box
- View the dashboard



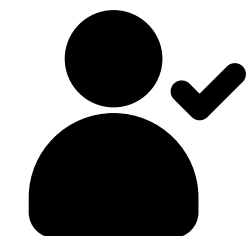
3. Main Actor and Features



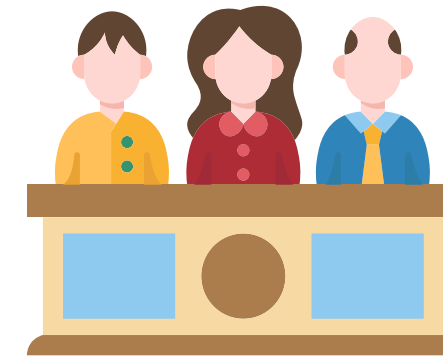
Coordinator

Feature:

- View informations of all events they have organized.
- Create an event.
- Management hosted the event.
- View teams, users assign to the event.
- View mentors, judges assigned to the event.
- Get logging about change infomation.
- View event's team submission.
- View progress of event's teams.
- View live score of the hosted event.



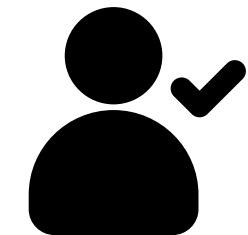
3. Main Actor and Features



Coordinator

Feature:

- View the dashboard.
- Get notification about unbalance scoring.
- Announce the results.
- View event's feedback



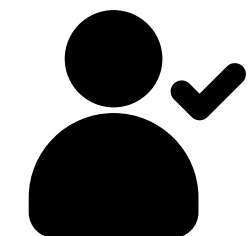
3. Main Actor and Features



Administrator

Feature:

- View audit logs
- Management events
- Management user
- View the dashboard
- View event's feedback
- Create an event.
- View teams, users assign to the event.
- View mentors, judges assigned to the event.



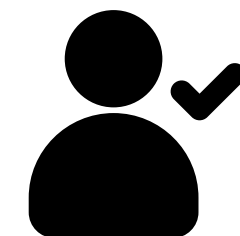
3. Main Actor and Features



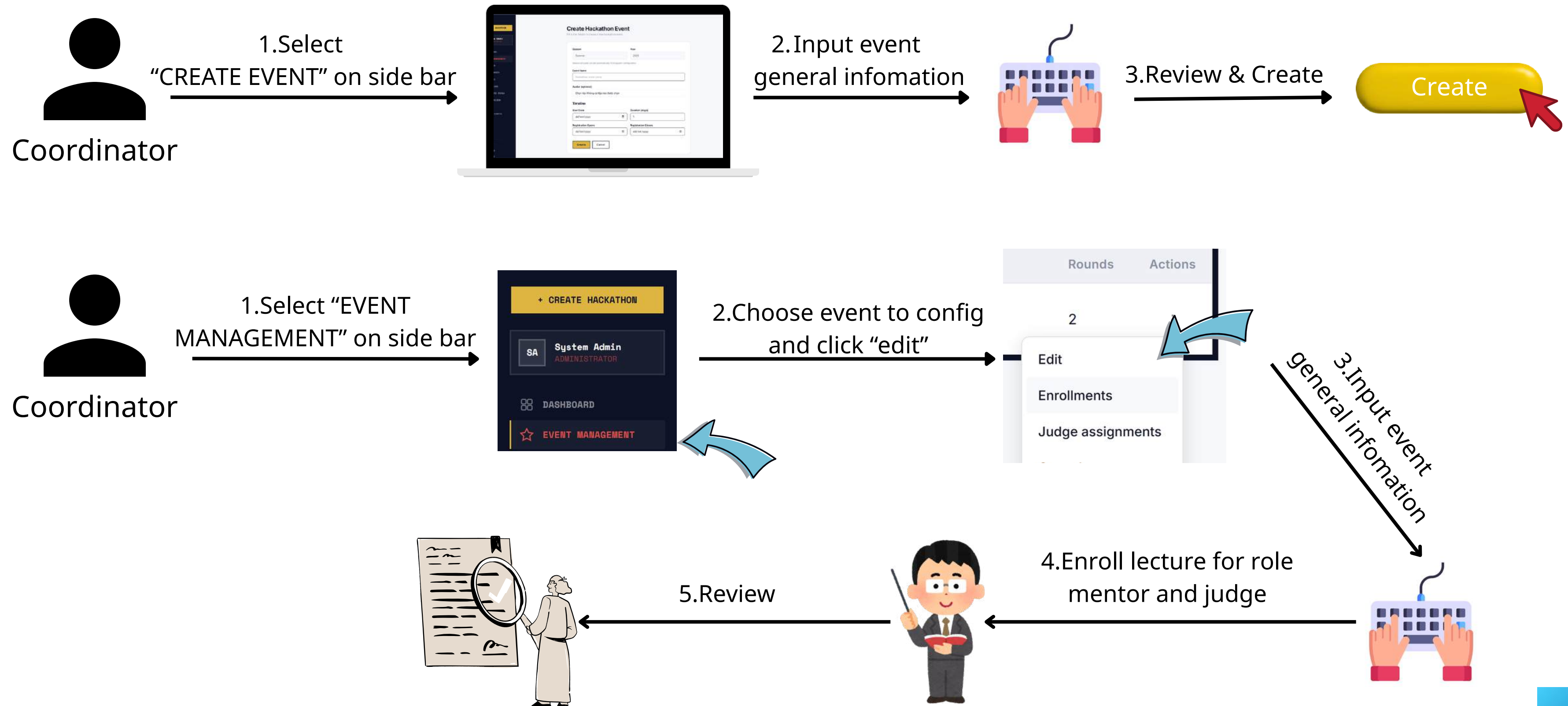
Administrator

Feature:

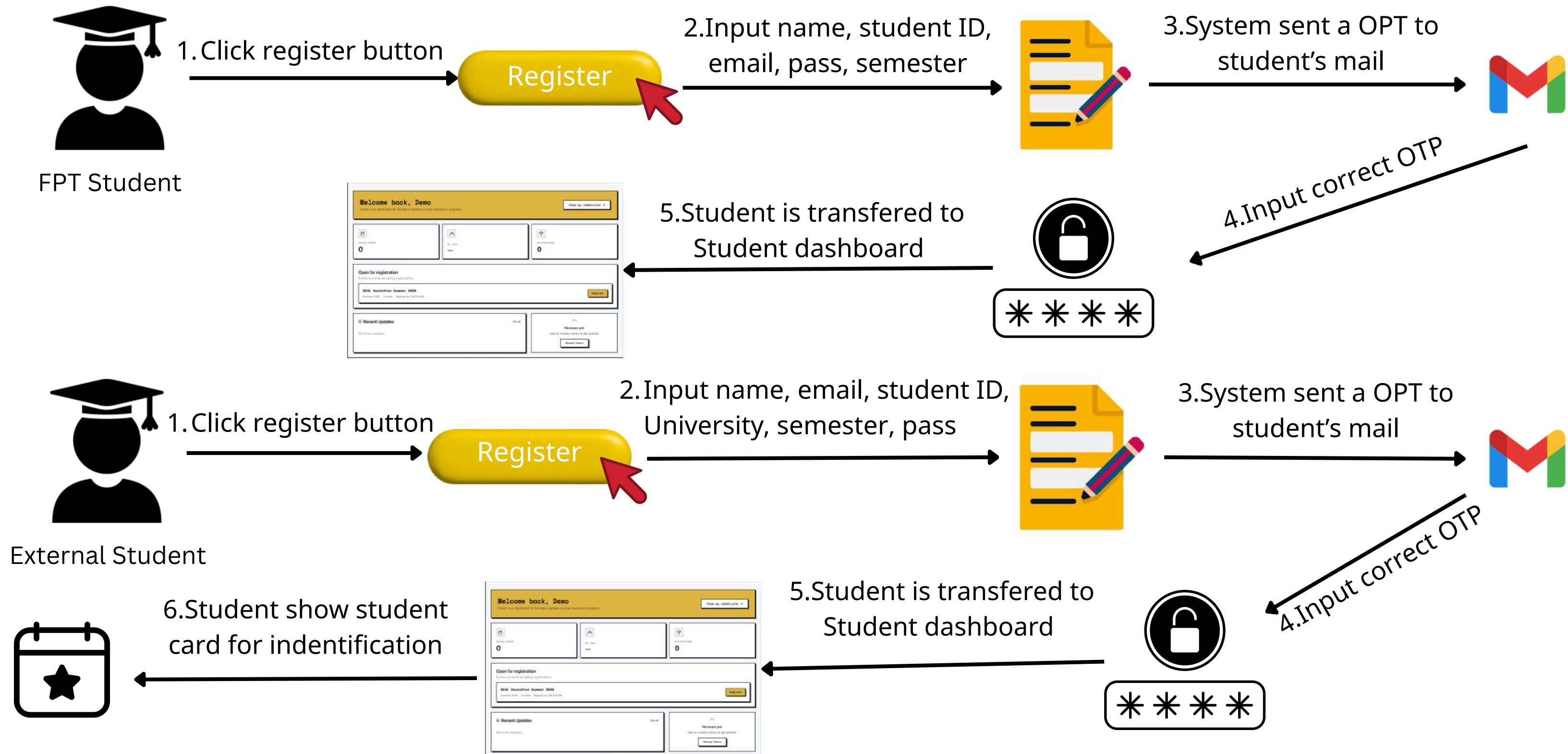
- View event's team submission.
- View progress of event's teams.
- View live score of the hosted event.
- Announce the results.



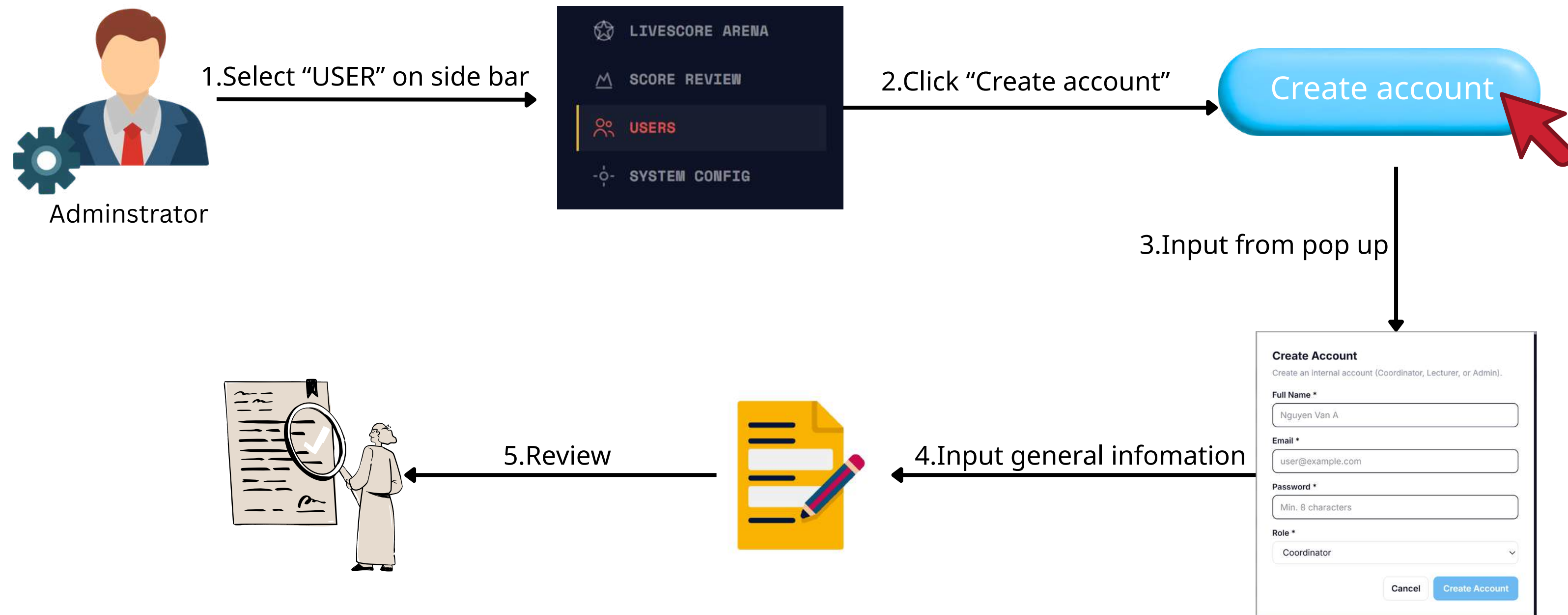
4. Flow | scenario 1: Create, Config Hackathon Event



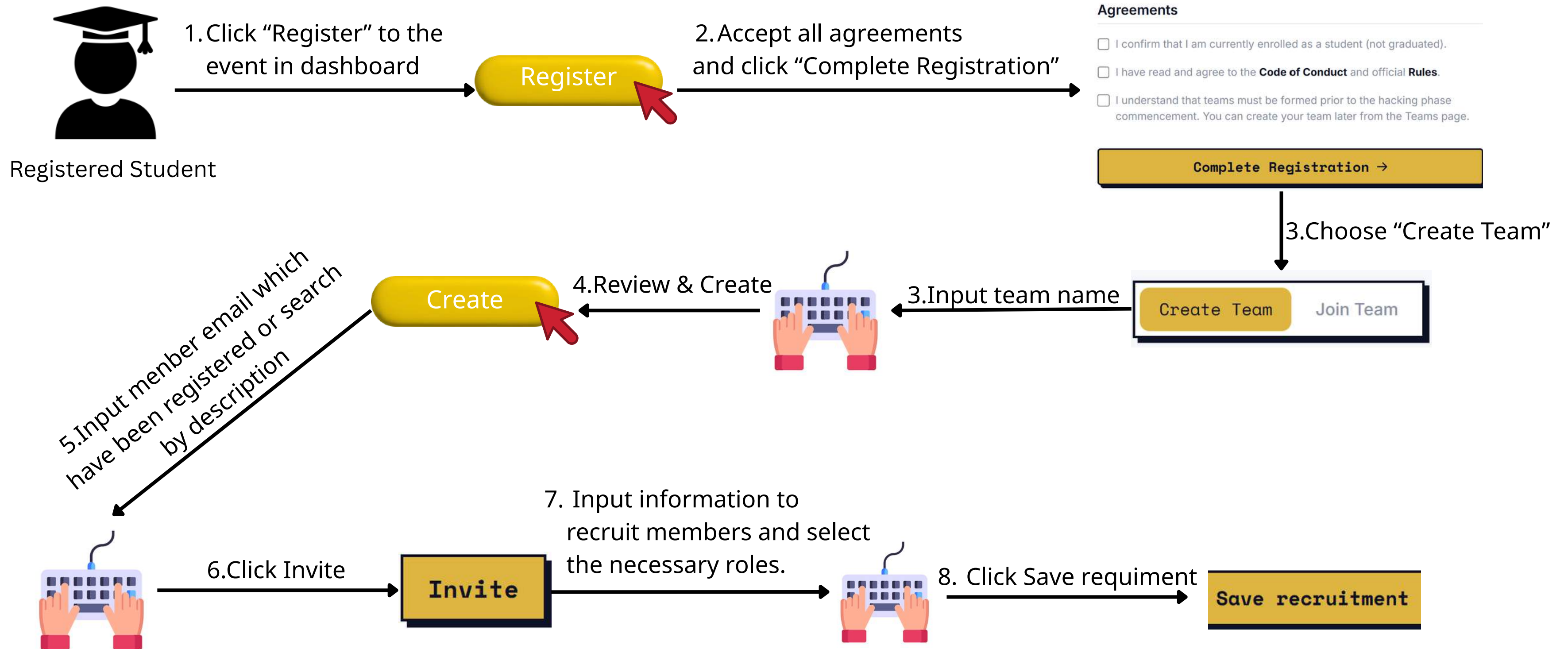
4. Flow | scenario 2: Registration & Verification FPT Student/External Student, Admin create lecture account



4. Flow | scenario 2: Admin (create lecture account)

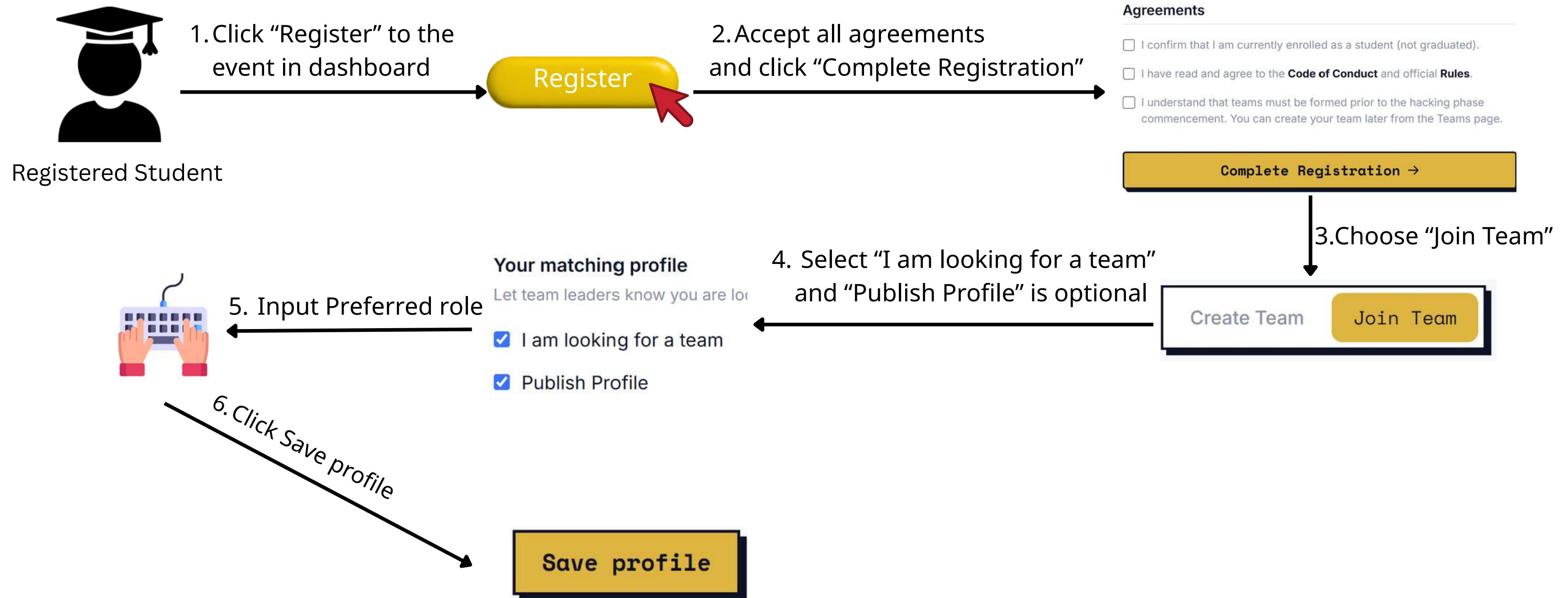


4. Flow | scenario 3: Team Formation



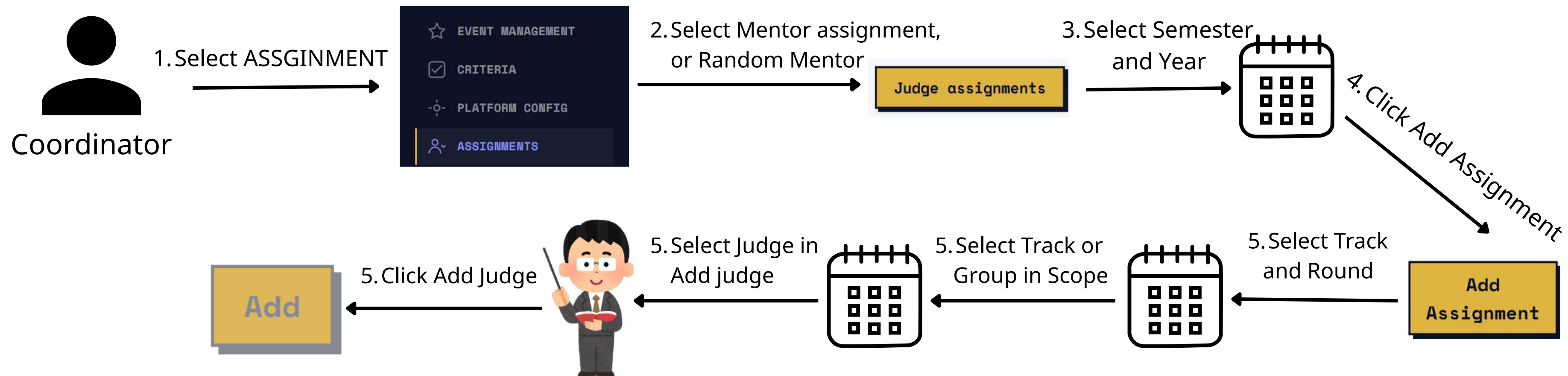


4. Flow | scenario 3: Team Formation



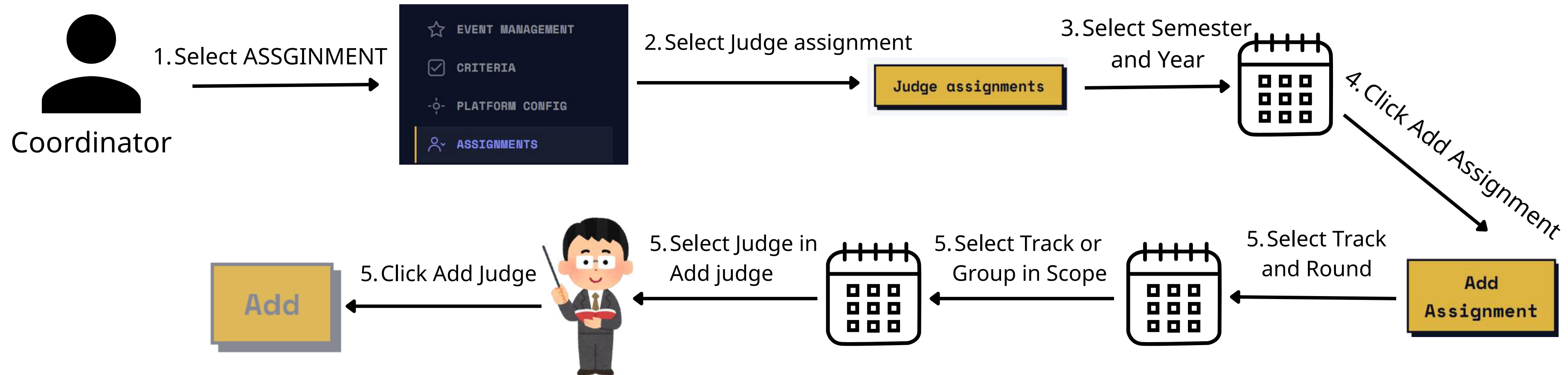
4. Flow | scenario 4: Competition Day (từ Assignment --> Submission --> Scoring)

Note: thêm 1 nhánh rút thăm bên ngoài ---> manual assign team



Làm theo thứ tự team assign trước → mentor → judge

4. Flow | scenario 4: Competition Day (từ Assignment --> Submission --> Scoring)





4. Flow | scenario 4: Competition Day (từ Assignment --> Submission --> Scoring)



4. Flow | scenario 5: Publish final result, Award, Feedback



Thank You For Listening

Q&A

